Project 2

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Summary

In this project we will consider a portfolio of non-life insurance contracts where n corresponds to the number of contracts in the porfolio. All of the contracts are assumed to be independent and generate claims during next year according to a Poission process with intensisty λ. One contract is assumed only to have a singelt payment per claim, however a contract can have more than one claim during a year.

Total claim cost distribution

The instructions is to simulate a portfolio 1000 times for the portfolio sizes n = 100, 1000 and 5000 with the parameter $\lambda = 0.1$, and then calculate the total (undiscounted) cost when the individual payemts are Gamma-distributed with mean 1 and standard deviation 0.2, and when the the individual payemts are Pareto-distributed with mean 1 and standard deviation 0.2. To be able to calculate the variance and the expected value for the total cost, we are using the following formula:

$$\mathbb{E}[S_N] = \mathbb{E}[N]\mathbb{E}[X_1]$$
$$Var(S_N) = \mathbb{E}[N]Var(X_1) + Var(N)E[X_1]$$

Where $N = \sum_{i=1}^{n} N_i \sim Po\left(\sum_{i=1}^{n} \bar{\lambda}_i\right)$ We can see the result in “Table 1” and “Table 2”, by analyzing the result we can see that the theoretical expected value when n = 100 is 10, n = 1000 is 100 and n = 5000 is 500. Comparing this values with the simulated we can see that they are not far away form what we calculated. The same thing if we look at the variance can be analyzed, the simulated variance is being close to what we calcultaded. By computing quantiles we can see from “Table 3” and “Table 4” that the 50%-quantile is close to the expected value, which we could have expected. If we now look if the collected data from the simulation have a normal distrubition we use “QQ-plots” which can be analyzed from watching “QQ 1-6”. We can see that when the claim size have an Pareto distribution or an Gamma distribution can be seen as having a normal distrubition when the amount of contracts is at least 1000, but with an amount of 5000 will give the best approximation.

Table 1: Expected total future cost, with Gamma dist. claim sizes compared with Pareto dist. claim sizes, with mean = 1 and sd = 0.2

	n = 100	n = 1000	n = 5000
$E[S_N]_{Gamma}$	9.962893	100.2094	499.8918
$E[S_N]_{Pareto}$	9.968091	100.0493	499.5244
$E[S_N]$	10.000000	100.0000	500.0000

Table 2: Variance total future cost, with Gamma dist. claim sizes compared with Pareto dist. claim sizes, with mean = 1 and sd = 0.2

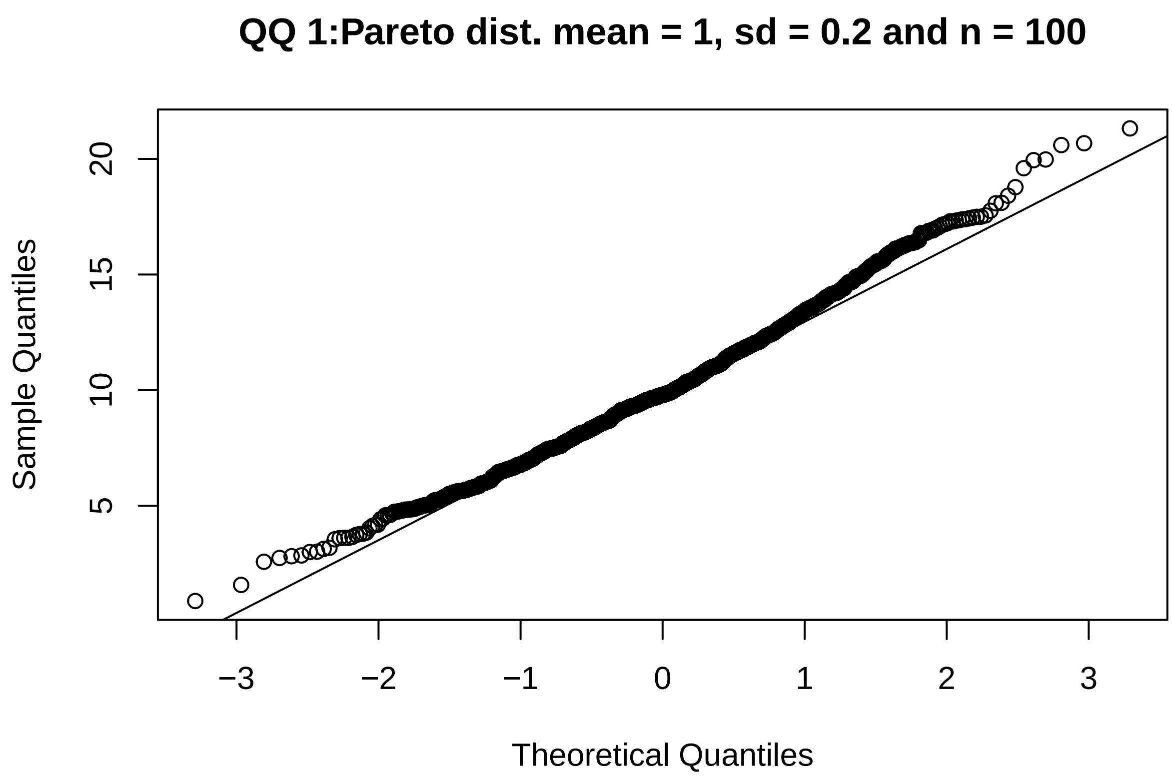
	n = 100	n = 1000	n = 5000
$Var(S_N)_{Gamma}$	10.49435	103.8422	520.1577
$Var(S_N)_{Pareto}$	10.32540	104.1982	518.9143
$Var(S_N)$	10.40000	104.0000	520.0000

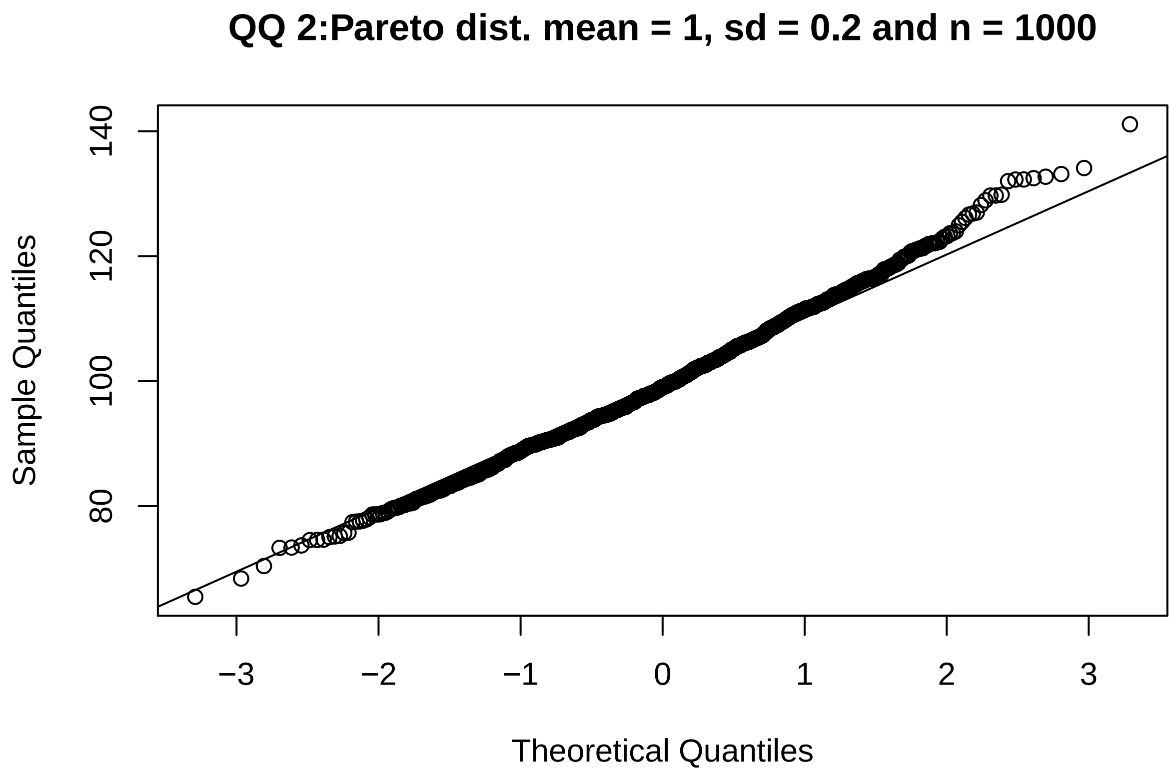
Table 3: Quantiles for Gamma dist. claim sizes with mean = 1 and sd = 0.2

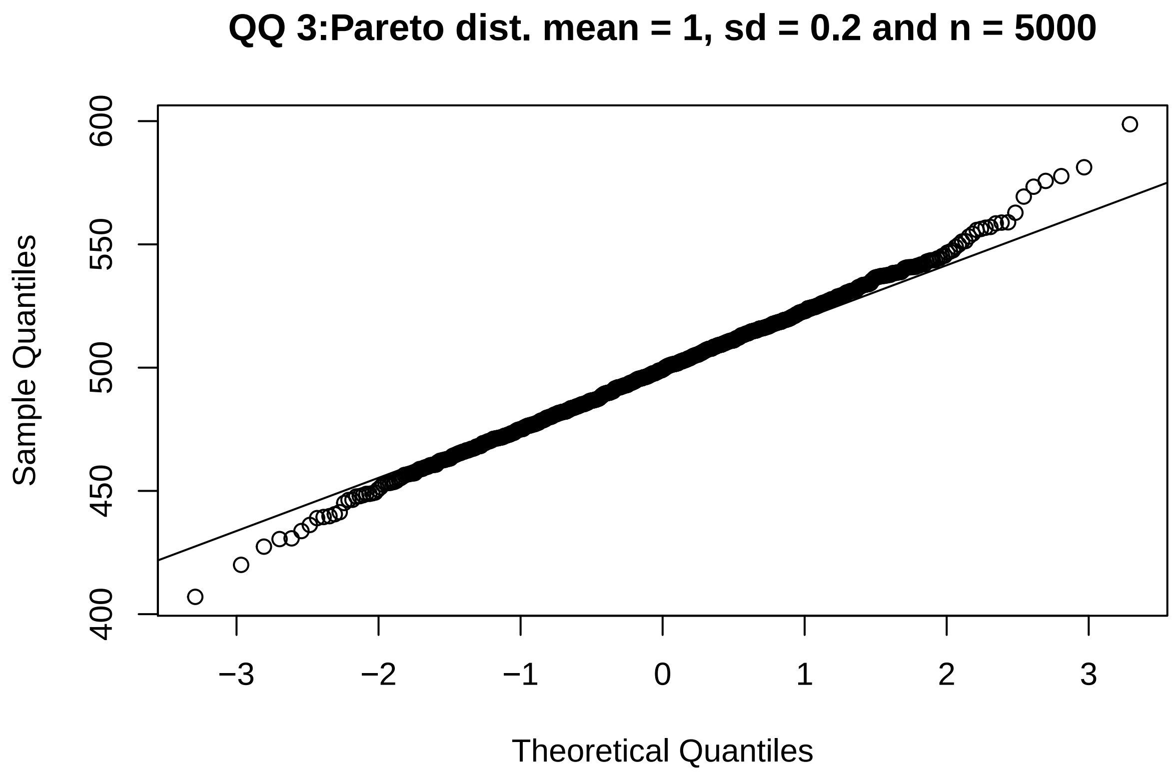
	n = 100	n = 1000	n = 5000
0%	1.843093	68.25654	419.6488
25%	7.642297	93.53366	484.6921
50%	9.679283	100.20084	499.7465
75%	11.867352	106.81612	515.3471
100%	23.596938	139.88641	568.3558

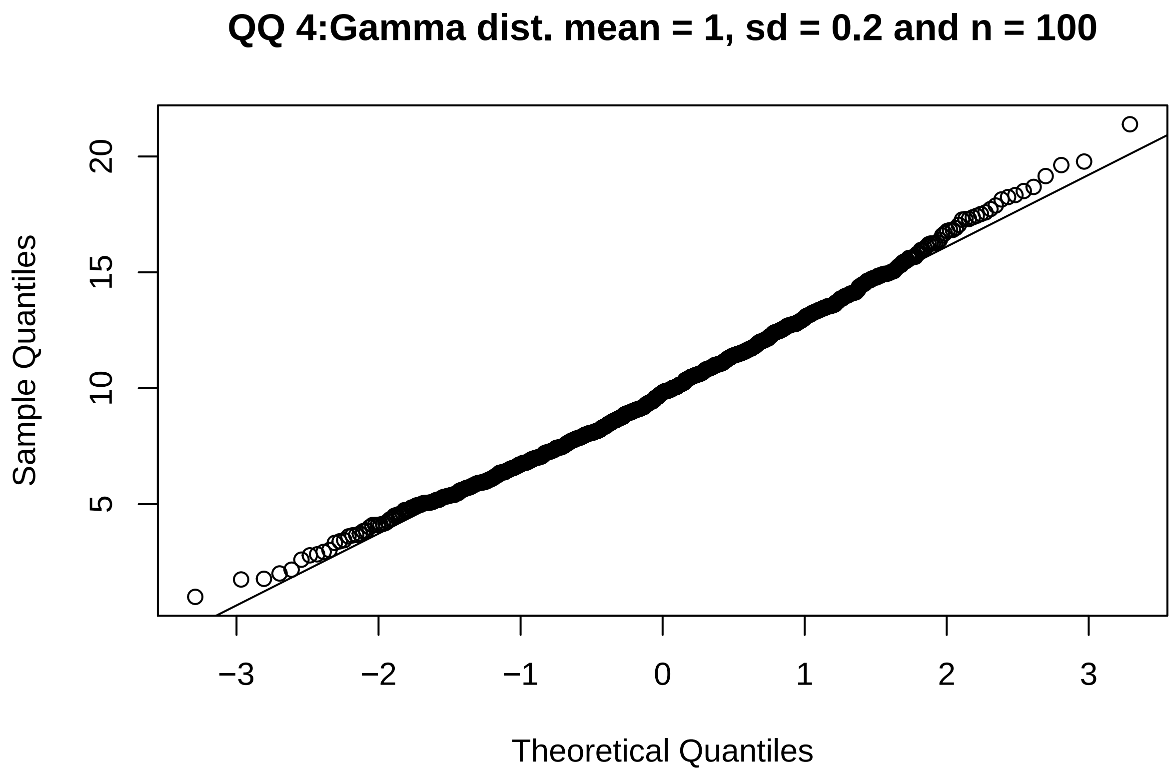
Table 4: Quantiles for Pareto dist. claim sizes with mean = 1 and sd = 0.2

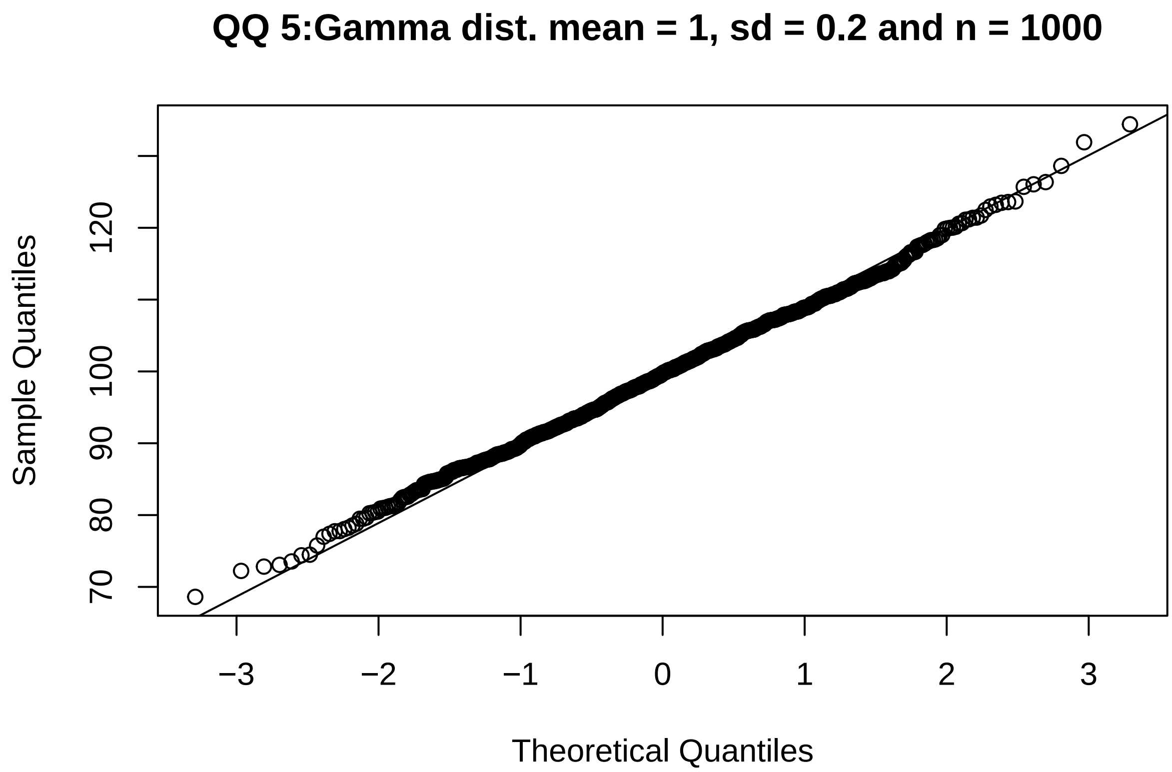
	n = 100	n = 1000	n = 5000
0%	0.000000	64.43017	431.6376
25%	7.899286	92.60079	483.4828
50%	10.074771	98.77528	500.9917
75%	12.308912	106.08840	516.1654
100%	22.509325	131.15355	576.7462

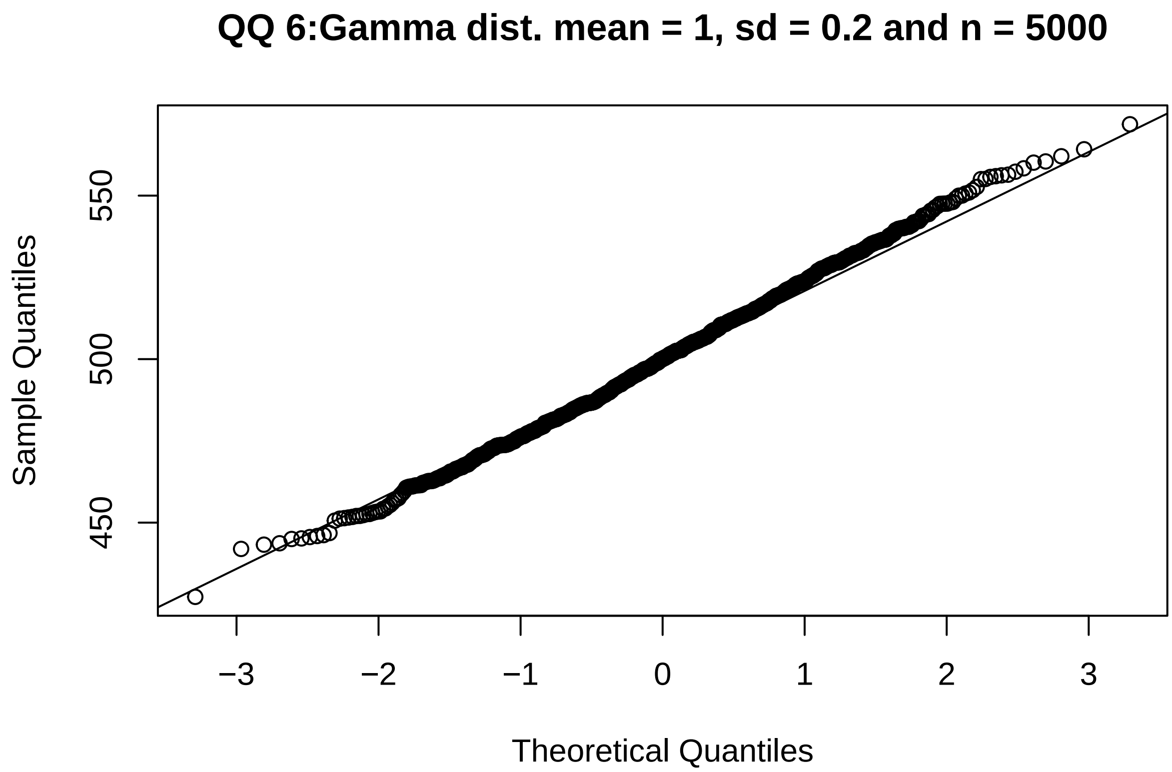



QQ 2:Pareto dist. mean = 1, sd = 0.2 and n = 1000
Sample Quantiles
80
100
120
140
−3
−2
−1
0
1
2
3
Theoretical Quantiles


QQ 3:Pareto dist. mean = 1, sd = 0.2 and n = 5000
Sample Quantiles
600
550
500
450
400
−3
−2
−1
0
1
2
3
Theoretical Quantiles


QQ 4:Gamma dist. mean = 1, sd = 0.2 and n = 100
Sample Quantiles
Theoretical Quantiles
5
10
15
20
−3
−2
−1
0
1
2
3


QQ 5:Gamma dist. mean = 1, sd = 0.2 and n = 1000
Sample Quantiles
70
80
90
100
120
−3
−2
−1
0
1
2
3
Theoretical Quantiles



If we now increase the standard devation to 0.5 for both the Gamma-distributed and Pareto-distributed claim sizes, we will get an different result, which can be obtaion by looking at "Table 5", "Table 6", "Table 7" and "Table 8", we can see that the total expected value hasn't change at all, but the variance has increased, for an example 10.4 to 12.5 when $n = 100$. By having a increased variance will imply that the quantiles will be diffrent from when the standrad deviation was 0.2. If we also analyze here if we can assume that the simulated data can be apporimated with an normal-distrubtion, we can see from "QQ 7-12" that same number of contract aslo applies here. A number of at least 1000 contracts will give the best approximation.

Table 5: Expected total future cost, with Gamma dist. claim sizes compared with Pareto dist. claim sizes, with mean = 1 and sd = 0.5

	n = 100	n = 1000	n = 5000
$E[S_N]_{Gamma}$	9.981679	101.0356	499.2780
$E[S_N]_{Pareto}$	9.903380	100.2686	500.5509
$E[S_N]$	10.000000	100.0000	500.0000

Table 6: Variance total future cost, with Gamma dist. claim sizes compared with Pareto dist. claim sizes, with mean = 1 and sd = 0.5

	n = 100	n = 1000	n = 5000
$Var(S_N)_{Gamma}$	12.56083	125.5973	626.8134
$Var(S_N)_{Pareto}$	12.20680	123.5770	623.7186

	n = 100	n = 1000	n = 5000
$Var(S_N)$	12.50000	125.0000	625.0000

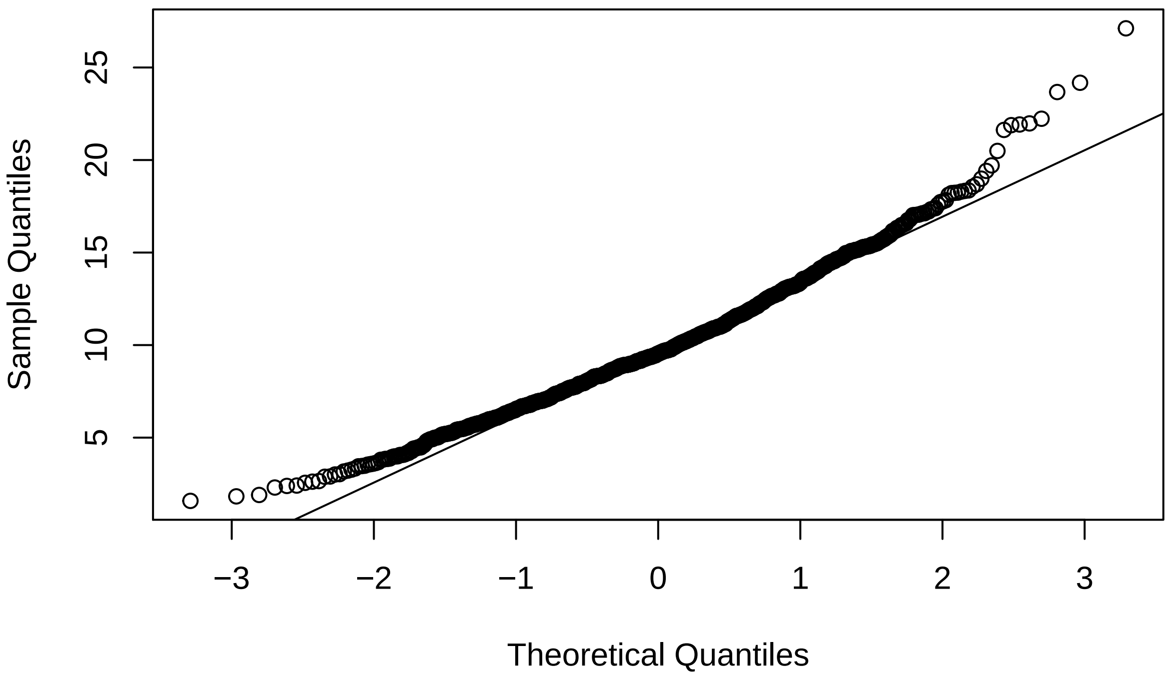
Table 7: Quantiles for Gamma dist. claim sizes with mean = 1 and sd = 0.5

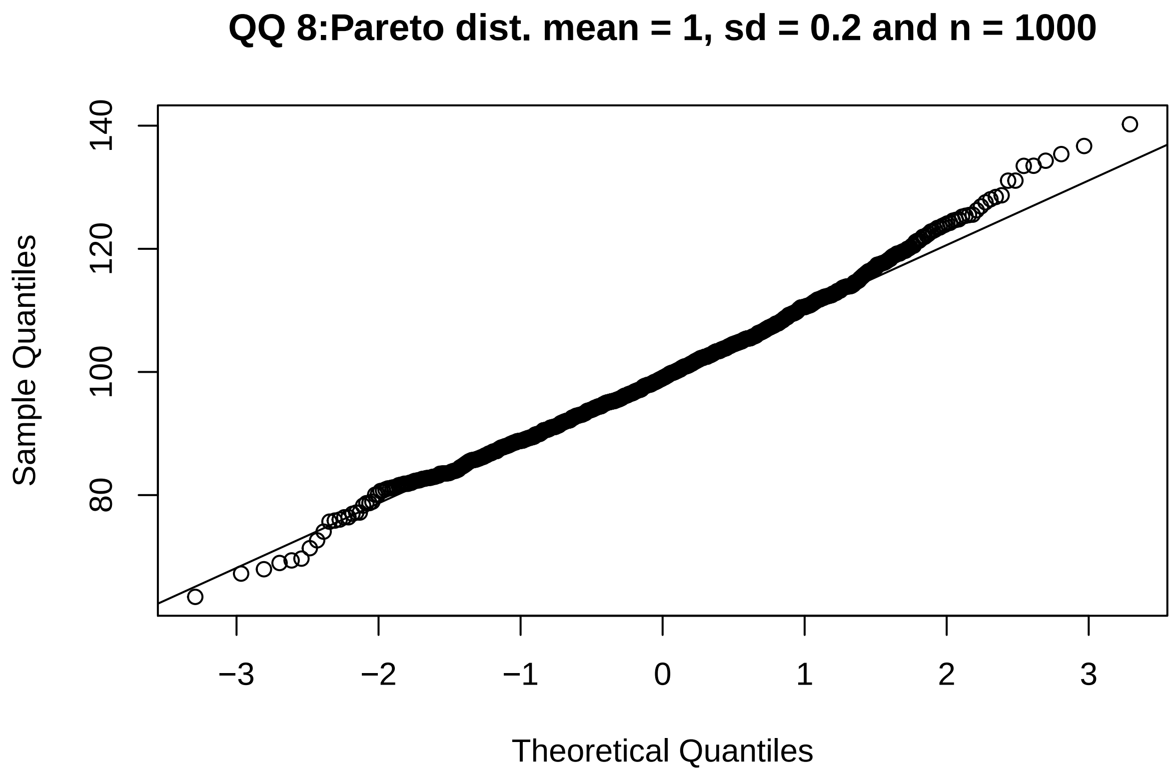
	n = 100	n = 1000	n = 5000
0%	1.602837	57.73135	418.0738
25%	7.559185	92.23602	483.2310
50%	9.791411	99.93626	499.8796
75%	12.168026	106.77727	516.7085
100%	21.167812	139.83888	585.9217

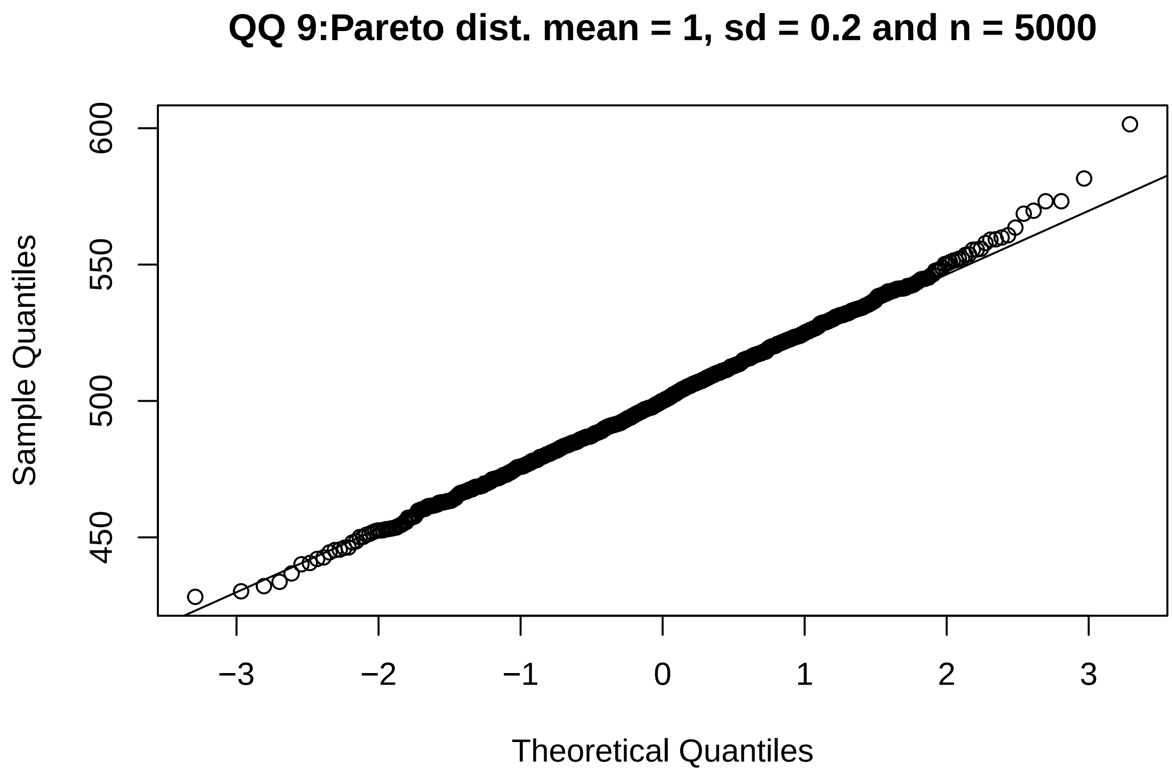
Table 8: Quantiles for Pareto dist. claim sizes with mean = 1 and sd = 0.5

	n = 100	n = 1000	n = 5000
0%	0.7791915	68.31564	431.0121
25%	7.6543440	92.17189	484.1481
50%	9.8483604	99.31281	500.0462
75%	12.0860090	106.71413	516.9558
100%	27.4600432	153.41093	591.2849

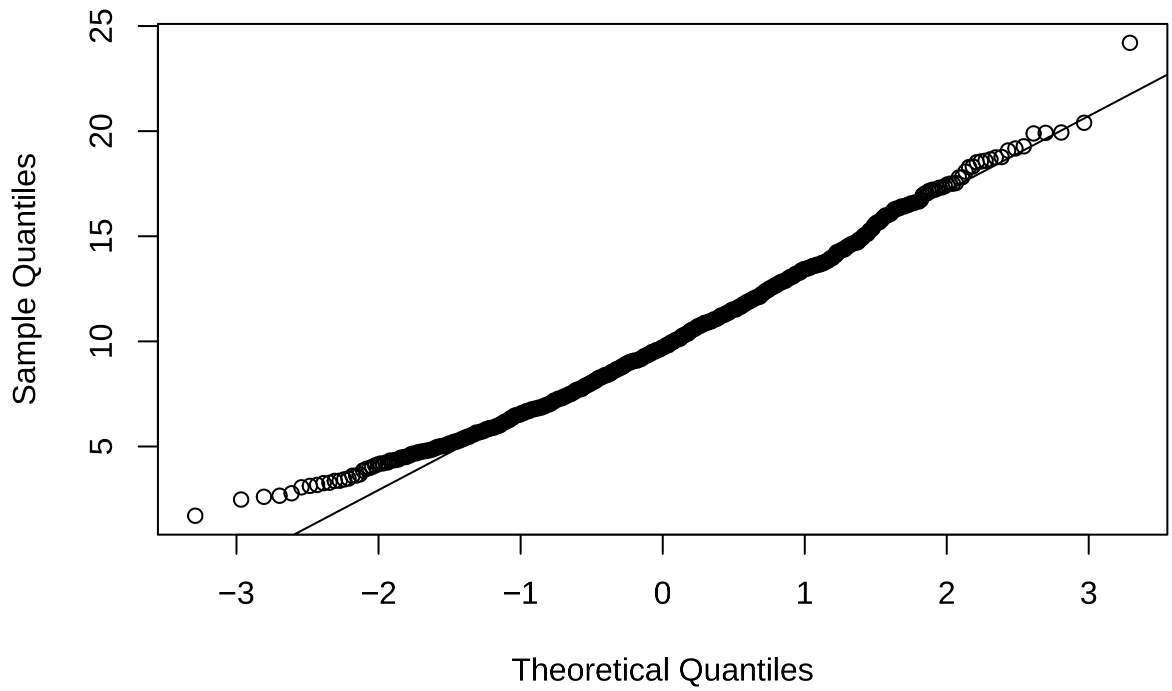
QQ 7:Pareto dist. mean = 1, sd = 0.2 and n = 100

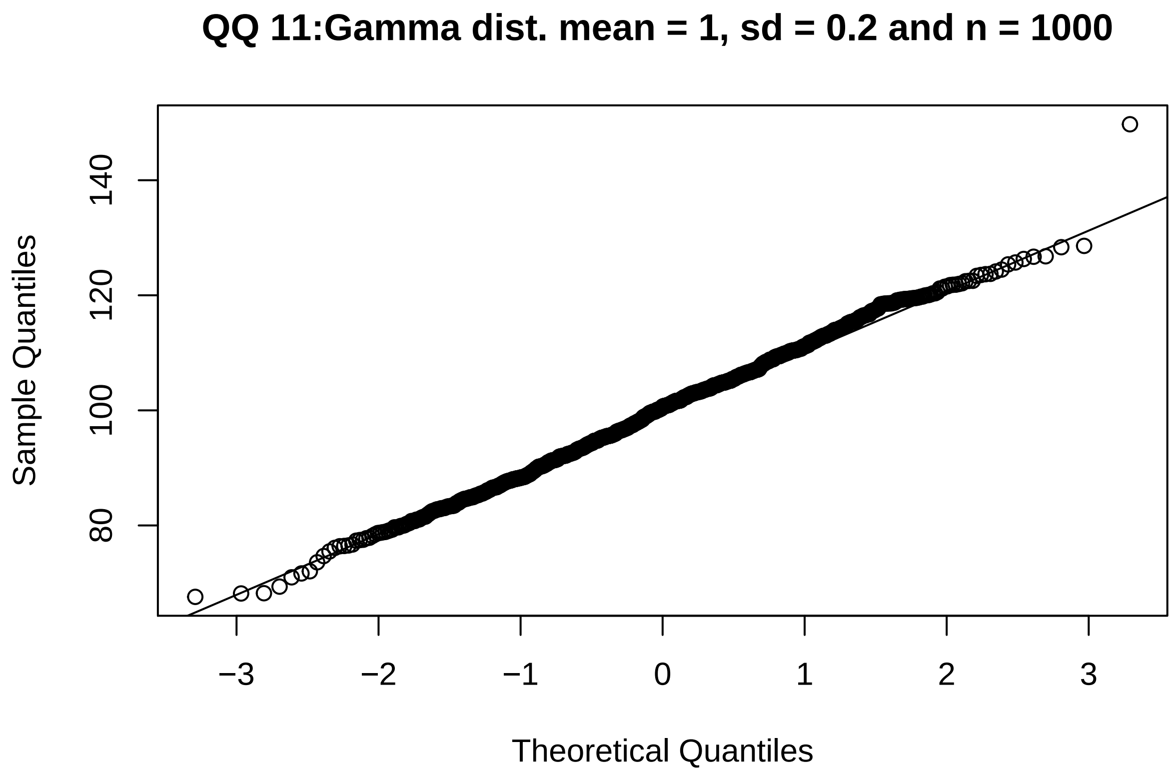


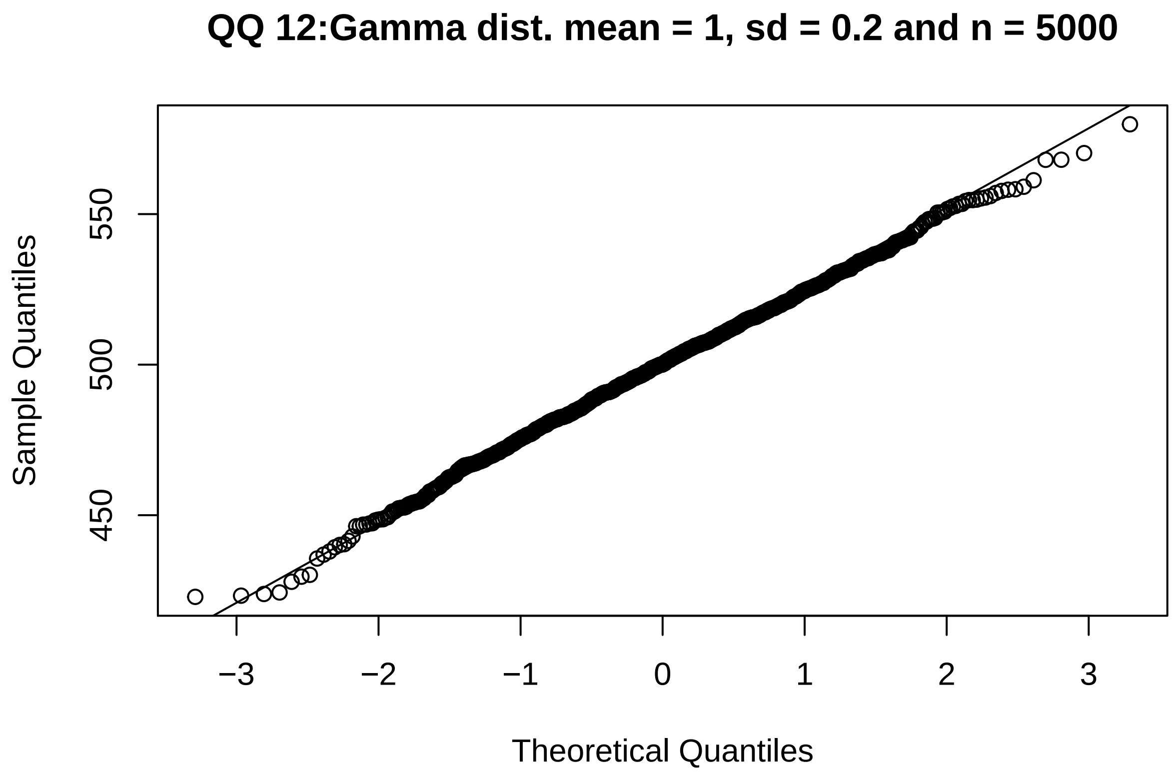

QQ 8:Pareto dist. mean = 1, sd = 0.2 and n = 1000
Sample Quantiles
140
120
100
80
−3
−2
−1
0
1
2
3
Theoretical Quantiles


QQ 9:Pareto dist. mean = 1, sd = 0.2 and n = 5000
Sample Quantiles
600
550
500
450
−3
−2
−1
0
1
2
3
Theoretical Quantiles

QQ 10:Gamma dist. mean = 1, sd = 0.2 and n = 100




QQ 11:Gamma dist. mean = 1, sd = 0.2 and n = 1000
Sample Quantiles
80
100
120
140
−3
−2
−1
0
1
2
3
Theoretical Quantiles



So far we have only looked at the total cost, but if we now focus on the total cost per contract instead. By simulate n contract individual for 1000 times, we can calculate the emperical mean, and as well the emperical 99%-percentile for each simulation. After collecting all the data we then calculte the mean and variance, and the result is listed in "Table 9-12". From this result we can see that already with only 100 contract the variance for the simulated total cost per contract, and the simulated 99%-percentile regardless which distribution the claim size have is trustworthy, but as in previously discussion a higher amout of contract will inreace the reliability of the result, and that conclusion can also be made from looking at "Table 9-12".

Table 9: (Pareto dist. with mean = 1 and sd =0.2.) The mean and variance of 1000 simulated mean of total cost per contract

	n = 100	n = 1000	n = 5000
Expected value	0.1005715	0.1002417	0.1000411
Variance	0.0009993	0.0001094	0.0000198

Table 10: (Gamma dist. with mean = 1 and sd =0.2.) The mean and variance of 1000 simulated mean of total cost per contract

	n = 100	n = 1000	n = 5000
Expected value	0.0994562	0.0996462	0.1000192
Variance	0.0011310	0.0001005	0.0000220

Table 11: (Pareto dist. with mean = 1 and sd =0.2.) The mean and variance of 1000 simulated 99%-percentiles of total cost per contract

	n = 100	n = 1000	n = 5000
Expected value	1.2437289	1.3367570	1.3331733
Variance	0.0783484	0.0203113	0.0037366

Table 12: (Gamma dist. with mean = 1 and sd =0.2.) The mean and variance of 1000 simulated 99%-percentiles of total cost per contract

	n = 100	n = 1000	n = 5000
Expected value	1.2731163	1.3276174	1.3288640
Variance	0.0524385	0.0056983	0.0010358

Claims Triangle

Observing the simulated values, further we present the following claims triangle. Each row represents a single *accident year* while each column represents observed *development years*, up to five years.

Table 13: Claims Triangle

Years	1	2	3	4	5
1	19.60	35.90	52.86	68.07	74.4
2	40.35	58.35	64.08	74.43	0.0
3	53.91	71.37	82.55	0.00	0.0
4	62.14	67.56	0.00	0.00	0.0
5	82.75	0.00	0.00	0.00	0.0

However, these observed values in (Table 1) corresponds to the first five years and that only. Seen from perhaps a clearer view, we observe (Figure 1), shown on the next page. Here we see that payments are made further than five years into the future, seen from the accident years. The following table (Table 2) shows the proportions of the payments made later then five years for each accident year.

Table 14: Claims Triangle

Accident.Years	Proportion of late payments
1	0.2008593
2	0.2925591
3	0.4544074
4	0.5479798
5	0.7005270

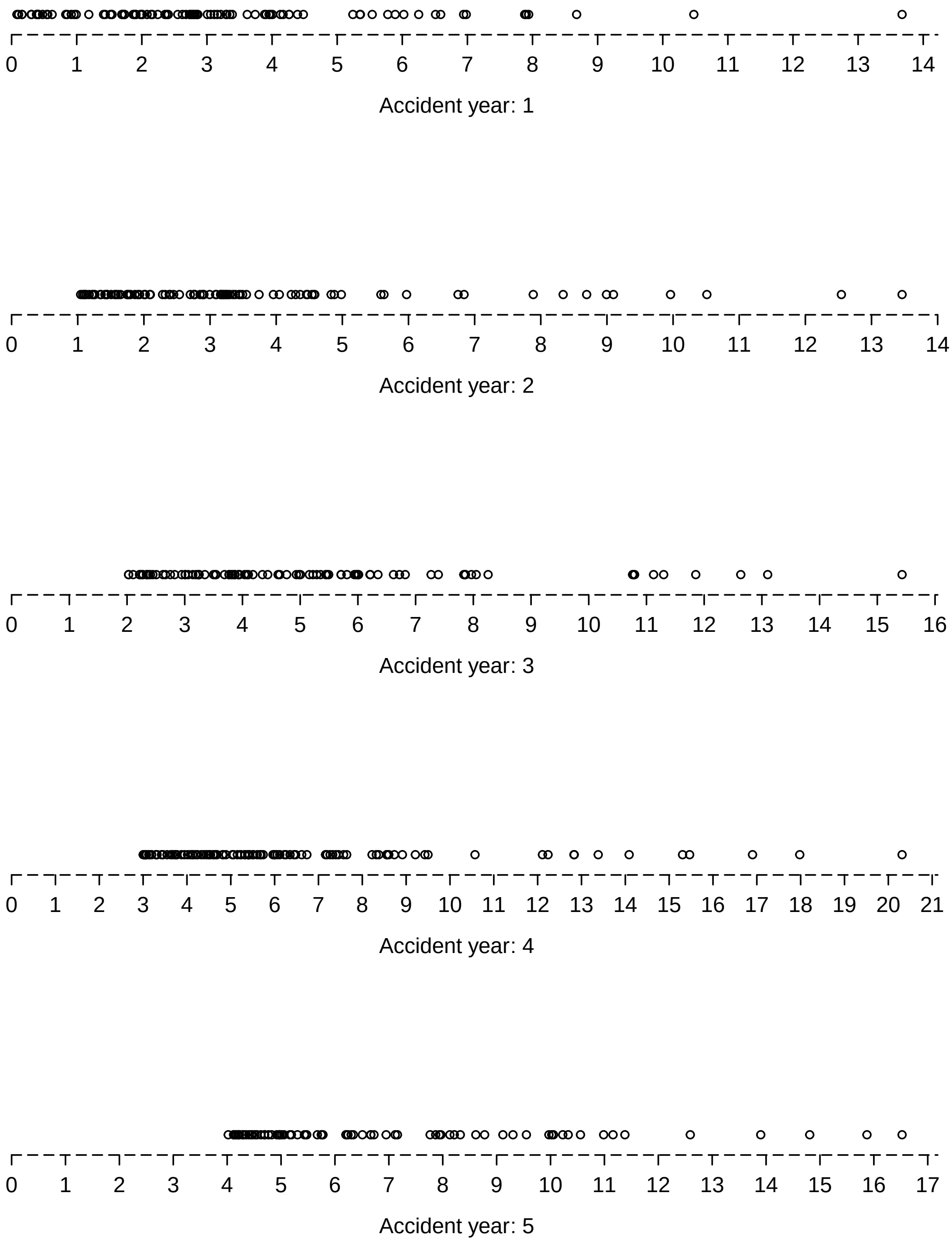


Figure 1: Timeline of accidents

We now observe the values given in (Table 1), which represents the claim triangle for the five given years. To further estimate the values in the lower-part of the triangle we will use the Chain-Ladder Method (CLM). Starting with finding the development factor to further estimate, we use the expression

$$f_k = \frac{\sum_{i=1}^{I-k} c_{i,k+1}}{\sum_{i=1}^{I-k} c_{i,k}},$$

where i represents the accident years $i = 1, 2, ..., I$, $I = 5$, and k represents the development years, $k = 1, 2..., I$. $c_{i,k}$ here is the value of a known total paid year i during development year k. f_k works as an estimation increase in the future values.
To now express the expected future value, we combine the development factor according to the following:

$$\mathbb{E}(C_{i,k+1}|C_{i,k}) = f_k C_{i,j},$$

the expected value of year i, development year $k+1$ equals the product of the value of year i, development year k and the development factor. From this, we get the following table:

Table 15: Claims Triangle - Using development factor

Years	1	2	3	4	5
1	19.6	35.9	52.86	68.07	74.4
2	40.35	58.35	64.08	74.43	81.13
3	53.91	71.37	82.55	100.71	109.77
4	62.14	67.56	81.41	99.32	108.26
5	82.75	109.23	131.62	160.58	175.03
D-factor		1.32	1.2	1.22	1.09

Table 16: Claims Triangle - Actual simulation

Years	1	2	3	4	5
1	19.60	35.90	52.86	68.07	74.40
2	40.35	58.35	64.08	74.43	78.39
3	53.91	71.37	82.55	86.14	89.43
4	62.14	67.56	72.51	77.91	81.09
5	82.75	85.89	88.74	92.47	93.89

These observed values now gives way for the calculation on the *Claims reserve*, noted as R_i for accident year i. We use the expression that

$$R_i = C_{i,I-i+1}(\Pi_{k=I-i+1}\hat{f}_k - 1),$$

given the last development year for accident year i, which corresponds to the cash flow, we can estime the remaining payments for the rest of the development years. Further, we now compare the calculated value of R_i with the actual simulated outcome:

Table 17: Claims reserve

I	Estimated Reserve	Simulated Reserve
2	7.30	3.96
3	33.21	6.88
4	48.50	13.53
5	120.85	11.14

We conclude by comparing the calculated development factor and the actual increasing factor, based on our simulations:

Table 18: Claims Triangle - Using development factor

Years	1	2	3	4	5
1		1.83	1.47	1.29	1.09
2		1.65	1.45	1.1	1.16
3		1.34	1.32	1.32	1.16
4		1.71	1.42	1.17	1.09
5		1.62	1.28	1.22	1.13
D-factor		1.32	1.2	1.22	1.09

Using likelihood-theory for right-censored data

To estimate the parameters in the Exp-distribution using likelihood-theory for right-censored data, we have the log-likelihood $l(\mu) = \sum_{i=1}^{n}(\delta)log(\alpha(t_i;\mu)) + log(S(t_i;\mu))$ knowing our particular $f(t;\mu)$ we can shorten $l(\mu) = dlog(\mu) + \tau\mu$ where $d = \sum_{i=1}^{n}\delta_i$ and $\tau = \sum_{i=1}^{n} t_i$ we then get $\frac{d}{d\mu}l(\mu) = \frac{d}{\mu} - \tau$ which gives $\tilde{\mu} = \frac{d}{\tau}$. $\delta_i = 1$ if it has been payed up by the time 5 or $\delta_i = 0$ if hasn't been payed. t_i will be the time it takes for the payment to be made or the time we have been waiting for the payment which then means $t_i = 5$

By only using the observed payments in your claims triangle

If we instead neglect that we know the number of payments, we have based on likelihood-theory $l(\mu) = \sum_{i=1}^{d} log(f(t_i;\mu)) = d\mu - \tilde{\tau}\mu$ where $d = \sum_{i=1}^{n}\delta_i$ and $\tilde{\tau} = \sum_{i=1}^{n}\delta_i t_i$ this gives us the estimator $\hat{\hat{\mu}} = \frac{d}{\tilde{\tau}}$. In this case we theoreticaly don't know if there will be a payment after the 5 year period which then means we can't add that to the $\sum t_i$ therefore $t_i = 0$ if $\delta_i = 0$. This will underestimate the mean compared to when we use the likelihood-theory for right-censored data.

Table 19: Parameter in the exp-distrubution

X.1	Likelihood-theory for right-censored data	Standard likelihood theory
Accidents year 0	0.3001449	0.519147
Accidents year 1	0.2261009	0.4116101
Accidents year 2	0.156684	0.2907649
Accidents year 3	0.1219701	0.2577675
Accidents year 4	0.05575915	0.2237472
Accidents for all 5 years	0.1459681	0.3314842